Big Data Analytics for Food Industry: Sales and Challenges

Anirudha Dhungana

Abstract— numerous industries, including the food business, are being revolutionized by big data analytics. Due to the abundance of data, the food industry has difficulties in supply chain optimization and decision-making. These problems are made more difficult by ineffective food waste management and a challenge in determining customer preferences and sales trends. The goal of this research is to improve strategy optimization and decision-making in the food business. This study detects trends in pizza sales, such as popular kinds, favored sizes, and ingredient utilization, by using data purification, preprocessing, modelling, visualization, and exploration approaches. The results support businesses in making defensible judgments by offering models for sales analysis and suggestions. By integrating big data analytics, the food industry can better match consumer requests, manage obstacles, and increase efficiency—all of which have a substantial positive impact on production and product quality.

Keywords—food sales, data cleaning, data visualization, Food industry, Big Data Analytics, PySpark, Logistic Regression, model

I. BACKGROUND OF THE STUDY

A. *General Information*

Big data states to the vast volume of organized and unstructured data that comes into a firm on a daily basis. The amount of data doesn't matter. Large-scale data collection, management, cleaning, and analysis are all part of big data analytics, which aids businesses in utilizing their big data.

Big data has a lot to offer the food business in terms of identifying patterns, the performance of new items, streamlining processes like delivery, forecasting sales, and even visualizing peak hours. Findings from the analysis of big data can improve operational and commercial choices. Big data allows businesses to integrate and analyses market data to identify unique enterprise kinds and generate new growth opportunities. These companies are able to compile and evaluate data regarding goods and services, customers and vendors, and consumer buying habits.

B. *Problem Statement*

Managing the enormous amount of data generated along the supply chain, from manufacturing to consumption, presents serious issues for the food business. Supply chain optimization is one major problem where big data analytics can be extremely important. By utilizing data, businesses can optimize transportation routes, reduce waste, and improve inventory management by improving demand predictions. Furthermore, in the food business, upholding safety and quality control requirements is crucial. Big data analytics can assist in tracking and evaluating data from several sources to spot irregularities, spot possible threats, and make sure legal obligations are being followed. By tackling these issues, the food sector may use big data to increase productivity, cut expenses, and eventually provide consumers with safer and better-quality products.

Figure 1: Food waste from different countries

Source: https://specials-images.forbesimg.com/imageserve/604200f81ee290bb440a68b2/960x0.jpg?fit=scale

C. *Aim and Objective*

When applying big data analytics to the food industry, this thorough paper intends to help organizations navigate issues including consumer data protection and privacy, data quality, dependability, and ethical considerations. It will carefully examine every aspect influencing food sales with the goal of improving sales results and spotting emerging trends. This research aims to enable businesses to succeed in the highly competitive food market by optimizing their strategy and utilizing data-driven insights to address important concerns.

D. *Contribution of the work connected with Methodology*

The analysis of food trends and business difficulties will be aided by this effort. By offering data on the foods that sell the best and the worst, it can also aid in reducing food waste. It may also highlight the difficulties with analytics in the food sector.

E. *Structure of the Report*

- Background of study
- Related works
- Methodology
- Result and Discussion
- Conclusion

II. RELATED WORK

The goal of Mihuandayani and colleagues' paper, "Food trend based on social media for big data analysis using K-mean clustering and SAW: A case study on Yogyakarta culinary industry," is to track consumer preferences by utilizing Twitter data that is pertinent to a particular area of interest. The data was unstructured because it was taken from social media. The authors make an effort to comprehend the importance of those data and extract only the details necessary to build a model that establishes meal rank. They also used Kmean Clustering and the Simple Additive Weighting approach to rank the foods. The accuracy of the model, which the authors assessed using data from four weeks, is seen when the result is compared with the available data.

In his work "Using Big Data and ML for Multilayered Observation for Healthy Food Environment and Diet," Fares Belkhiria used census data to build seven models—Naïve Bayes, Decision Tree, Random Forest, KNearest Neighbor, Support Vector Machine, XGBoost, and Neural Network—that can forecast the state of the food environment. Thirty percent went towards testing, while the remaining seventy percent was put to use for instruction. Models such as Random Forest, K-Nearest Neighbor, and Decision Tree outperform other models when all models are examined to an accuracy of 60%. In the end, the authors draw the conclusion that adding more census data to the models' training would improve their accuracy.

This paper big data in food supply chain management looks at how big data analytics affects the efficiency of the food industry's supply chain and its financial gains. The author explain to the readers the intriguing nature of the following: they may enhance supply chain visibility, forecast or predict demand, and optimize inventory levels, in that order, by aggregating and analyzing data from multiple sources, including sensors, RFID tags, and social media. This paper claims that big data analysis has already been used in a number of cases that have increased customer satisfaction and resulted in significant cost savings.

The paper Predictive Analytics for Food Industry Sales Forecasting: The application of predictive analytics methods to food industry sales forecasting is the main topic of this study. The accuracy and dependability of several models, like as regression models, time series analysis, and machine learning algorithms, are assessed in terms of their capacity to forecast sales patterns. The authors show how adding outside data—like social media mood, economic indicators, and weather patterns—can improve sales prediction accuracy and facilitate more informed resource allocation and decision-making.

This paper Challenges and Opportunities of Implementing Big Data Solutions in Food Retailing: looks at the particular difficulties food retailers have when implementing big data analytics. There includes discussion of topics like data privacy, integrating different data sources, and the necessity of real-time analytics. The study also looks at the potential that big data analytics offers, including enhanced customer experience, dynamic pricing, and tailored marketing. The authors offer a methodology for resolving these issues and using big data to gain an edge over competitors in the retail food industry.

This review of The Role of Big Data in Enhancing Food Safety and Quality Control: looks into how big data analytics can improve quality control and food safety. The writers go through a number of data sources, such as sensor data from manufacturing lines, inspection logs, and customer reviews. The potential of analytical approaches including anomaly detection, predictive maintenance, and real-time monitoring to avert food safety mishaps and guarantee constant quality is investigated. There are case studies of effective implementations in several food industry segments provided.

This paper Big Data Analytics for Customer Behavior Analysis in the Food Industry: explores how big data analytics can be used to comprehend and forecast consumer behavior in the food business. Businesses can learn about consumer preferences, purchasing trends, and satisfaction levels by examining huge datasets from sources like loyalty programs, online reviews, and social media interactions. The study highlights cutting-edge methods that aid in customizing marketing campaigns, boosting client retention, and improving the entire consumer experience, such as sentiment analysis, clustering, and recommendation systems.

III. MEHODOLOGY

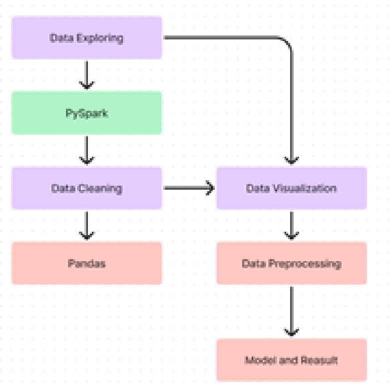

Figure 2: Methodology

1. Data Exploring

The dataset was acquired from Kaggle, a platform that provides a variety of data that is still pending analysis. Like the prior Kaggle dataset, this one has been cleaned and arranged. There are 12 columns and 48620 rows of data in this dataset. Order number, order details number, pizza number, quantity, order date, order time, unit price, total price, pizza size, pizza category, pizza ingredients, and pizza name are some of the details it contains. This project employs tools such as PySpark for data analysis in order to determine the category, sort of pizza, most sales, and overall sales.

2. Data Cleaning

Since the majority of the Kaggle datasets have been cleaned, any null values in the dataset can be filled in with average values, or any column with a significant number of null values that is not necessary for the model or visualization can be dropped. This project uses the Google collab platform for preprocessing and model building, PySpark to read the do the task from CSV file.

3. Data Preprocessing

Utilizing PySpark to load and read the data. One hot encoding is used to convert categorical values into numerical values because the model is unable to handle them. The dataset is then divided into 80 to 20 ratios if our model needs to use them for training and testing.

4. Models

These are the models that can be constructed with the available data because the dataset which is used in this study is about pizza sales. The components Analysis Model can assist businesses in negotiating ingredient costs with associated suppliers by analyzing pizza ingredient data to identify the top 10 components used in pizza preparation. To help restaurants better understand the demand for pizza, another way may be to identify the pizza that is most and least sold. Another recommendation model that takes the name of a pizza and checks other pizza with their comparable ingredients and suggests it to users may be developed by using the pizza's name and pizza's ingredient.

5. Data Visualization

This project or system practices PySpark, which is used to analyze data using pie charts, line charts, or even bar charts, is used for data visualization. Even though this project creates a model to identify the best pizza. The visual representation helps to make the data easily interpretable.

IV. REASULT AND DISCUSSION

In PySpark, data exploration involves loading data into a DataFrame, checking its structure and summary statistics, cleaning any missing or erroneous data, performing transformations like filtering or aggregating, and visualizing insights for analysis.

a. Data Visualization

```
root
 |-- pizza_id: double (nullable = true)
 |-- order_id: double (nullable = true)
 |-- pizza_name_id: string (nullable = true)
 |-- quantity: double (nullable = true)
 |-- order_date: date (nullable = true)
 |-- order_time: timestamp (nullable = true)
 |-- unit_price: double (nullable = true)
 |-- total_price: double (nullable = true)
 |-- pizza_size: string (nullable = true)
 |-- pizza_category: string (nullable = true)
 |-- pizza_ingredients: string (nullable = true)
 |-- pizza_name: string (nullable = true)
 |-- year: integer (nullable = true)
 |-- month: integer (nullable = true)
 |-- day: integer (nullable = true)
 |-- hour: integer (nullable = true)
 |-- minute: integer (nullable = true)
```

Figure 3: Data Visualization

In the above picture, there are the columns that are in the datasets with their data type.

pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name
1.0	1.0	hawaiian_m	1.0	1/1/2015	11:38:36	13.25	13.25	M	Classic	Sliced Ham, Pineapple, Mozzarella Cheese	The Hawaiian Pizza
2.0	2.0	classic_dlx_m	1.0	1/1/2015	11:57:40	16.0	16.0	M	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Bacon	The Classic Deluxe Pizza
3.0	2.0	five_cheese_l	1.0	1/1/2015	11:57:40	18.5	18.5	L	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Gouda Cheese, Romano Cheese, Blue Cheese, Garlic	The Five Cheese Pizza
4.0	2.0	ital_supr_l	1.0	1/1/2015	11:57:40	20.75	20.75	L	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
5.0	2.0	mexicana_m	1.0	1/1/2015	11:57:40	16.0	16.0	M	Veggie	Tomatoes, Red Peppers, Jalapeno Peppers, Red Onions, Cilantro, Corn, Chipotle Sauce, Garlic	The Mexicana Pizza
6.0	2.0	thai_ckn_l	1.0	1/1/2015	11:57:40	20.75	20.75	L	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, Thai Sweet Chilli Sauce	The Thai Chicken Pizza
7.0	3.0	ital_supr_m	1.0	1/1/2015	12:12:28	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
8.0	3.0	prsc_argla_l	1.0	1/1/2015	12:12:28	20.75	20.75	L	Supreme	Prosciutto di San Daniele, Arugula, Mozzarella Cheese	The Prosciutto and Arugula Pizza
9.0	4.0	ital_supr_m	1.0	1/1/2015	12:16:31	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
10.0	5.0	ital_supr_m	1.0	1/1/2015	12:21:30	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
11.0	6.0	bbq_ckn_s	1.0	1/1/2015	12:29:36	12.75	12.75	S	Chicken	Barbecued Chicken, Red Peppers, Green Peppers, Tomatoes, Red Onions, Barbecue Sauce	The Barbecue Chicken Pizza
12.0	6.0	the_greek_s	1.0	1/1/2015	12:29:36	12.0	12.0	S	Classic	Kalamata Olives, Feta Cheese, Tomatoes, Garlic, Beef Chuck Roast, Red Onions	The Greek Pizza
13.0	7.0	spinach_supr_s	1.0	1/1/2015	12:50:37	12.5	12.5	S	Supreme	Spinach, Red Onions, Pepperoni, Tomatoes, Artichokes, Kalamata Olives, Garlic, Asiago Cheese	The Spinach Supreme Pizza
14.0	8.0	spinach_supr_s	1.0	1/1/2015	12:51:37	12.5	12.5	S	Supreme	Spinach, Red Onions, Pepperoni, Tomatoes, Artichokes, Kalamata Olives, Garlic, Asiago Cheese	The Spinach Supreme Pizza
15.0	9.0	classic_dlx_s	1.0	1/1/2015	12:52:01	12.0	12.0	S	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Bacon	The Classic Deluxe Pizza
16.0	9.0	green_garden_s	1.0	1/1/2015	12:52:01	12.0	12.0	S	Veggie	Spinach, Mushrooms, Tomatoes, Green Olives, Feta Cheese	The Green Garden Pizza
17.0	9.0	ital_cpcllo_l	1.0	1/1/2015	12:52:01	20.5	20.5	L	Classic	Capocollo, Red Peppers, Tomatoes, Goat Cheese, Garlic, Oregano	The Italian Capocollo Pizza
18.0	9.0	ital_supr_l	1.0	1/1/2015	12:52:01	20.75	20.75	L	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
19.0	9.0	ital_supr_s	1.0	1/1/2015	12:52:01	12.5	12.5	S	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
20.0	9.0	mexicana_s	1.0	1/1/2015	12:52:01	12.0	12.0	S	Veggie	Tomatoes, Red Peppers, Jalapeno Peppers, Red Onions, Cilantro, Corn, Chipotle Sauce, Garlic	The Mexicana Pizza

only showing top 20 rows

Figure 4: Top 20 pizza details from the dataset

In the above picture, it's the data frame in which the top 20 information are displayed.

Pizza Size	Sold Percentage
XXL	0.06%
XL	1.12%
Big	38.12%
Medium	31.53%
Small	29.08%

The above table displays the total number of pizza sales according to size. This data helps to make a plan to focus on which size.

Pizza Type	Sales Percentage
Classic	30.13%
Supreme	24.08%
Veggie	23.10%
Chicken	22.19%

According to the above table, Classic pizza is the most popular type of pizza, selling for 30.03% of the total, while Chicken pizza has a lesser share.

Figure 5: Tree map according to pizza sales and category

The above figure helps to understand as which category pizza can be most sold.

The major problem we have discussed about is the wastage problem, if the restaurant or anyone gets the information about the most used and less used s/he will get the ingredients accordingly.

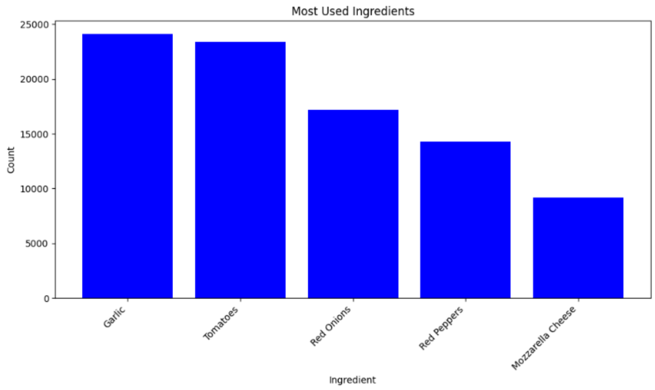

Figure 6: Most Used ingredients in pizza

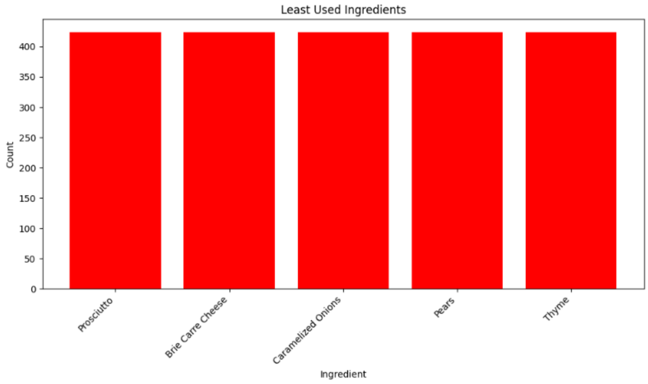

Figure 7: Least used ingredients in pizza

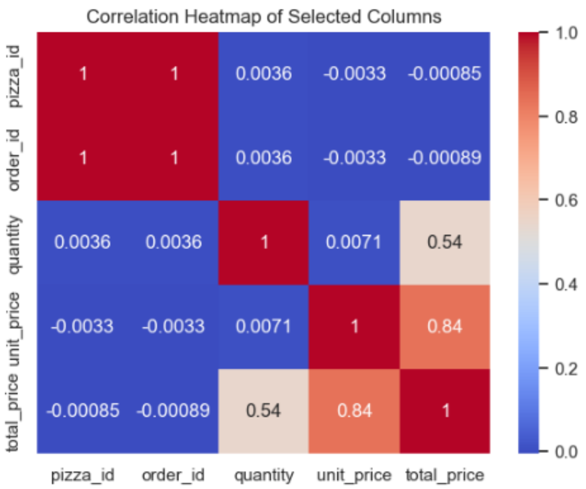

Figure 8: Correlation heat map

The above figure, each cell in a correlation heat map indicates the correlation coefficient between two variables, providing a visual representation of the relationship between the variables in a dataset.

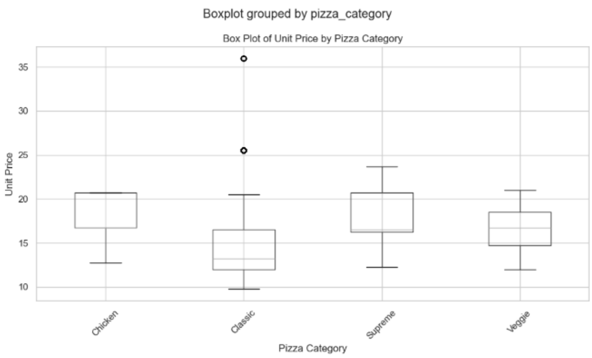

Figure 9: Box plot

In the above figure, it facilitates the comparison of distributions of variables or groups, the detection of outliers, and the determination of the data's central tendency and spread.

1. Recommended Model

The pizza name and dataset are entered into this model, which then extracts all the elements needed to construct the inputted pizza and finds a recommendation for a different pizza with comparable ingredients using Logistic Regression. For an ingredient to be on the recommended list, there must be three or more similarities between the substances. The model is denoted by the following figure:

```
def get_recommendations(pizza_name, pizza_df):
    input_ingredients = set(pizza_df.filter(pizza_df['pizza_name'] == pizza_name)
                                    .select('pizza_ingredients')
                                    .collect()[0]['pizza_ingredients'].split(', '))
    print(f"Ingredients of {pizza_name}: {', '.join(input_ingredients)}")
    similar_pizzas = set()  # Using a set to store unique recommended pizzas
    similar_pizzas_ingredients = {}
    for row in pizza_df.collect():
        if row['pizza_name'] != pizza_name:
            pizza_ingredients = set(row['pizza_ingredients'].split(', '))
            common_ingredients_temp = input_ingredients.intersection(pizza_ingredients)
            if len(common_ingredients_temp) >= 3:
                similar_pizzas.add(row['pizza_name'])  # Adding recommended pizza to the set
                similar_pizzas_ingredients[row['pizza_name']] = pizza_ingredients
    return list(similar_pizzas), similar_pizzas_ingredients

# Test the recommendation system
recommended_pizzas, recommended_pizzas_ingredients = get_recommendations('The California Chicken Pizza', df)
print('Recommended Pizzas:', recommended_pizzas)
print('Ingredients of Recommended Pizzas:')
for pizza in recommended_pizzas:
    print(pizza, ':', ', '.join(recommended_pizzas_ingredients[pizza]))
```

```
Ingredients of The California Chicken Pizza: Artichoke, Gouda Cheese, Garlic, Fontina Cheese, Chicken, Jalapeno Peppers, Spinach
Recommended Pizzas: ['The Chicken Pesto Pizza']
Ingredients of Recommended Pizzas:
The Chicken Pesto Pizza : Tomatoes, Garlic, Pesto Sauce, Chicken, Spinach, Red Peppers
```

Figure 10: Recommended Model

2. To get best and worst pizza by sales

```
# Group the data by pizza name and sum the total quantity sold
pizza_sales = df.groupBy('pizza_name').agg(F.sum('quantity').alias('total_quantity_sold'))

# Find the highest and lowest selling pizzas
best_selling_pizza = pizza_sales.orderBy(F.desc('total_quantity_sold')).first()
worst_selling_pizza = pizza_sales.orderBy('total_quantity_sold').first()

# Print the results
print(f"The best selling pizza is {best_selling_pizza['pizza_name']}, with a total of {best_selling_pizza[
print(f"The worst selling pizza is {worst_selling_pizza['pizza_name']}, with a total of {worst_selling_piz
```

```
The best selling pizza is The California Chicken Pizza, with a total of 14574.0 units sold.
The worst selling pizza is The Brie Carre Pizza, with a total of 2598.0 units sold.
```

Figure 11: Best and worst pizza according to sales

This model determines the best and worst-selling pizza based on the number and name of the pizza.

Here, the model performs really well every time I give a different pizza name, different suggestion and the ingredients the give name pizza is using and the suggested pizza will use will be displayed.

V. CONCLUSION

For giants in the food business, integrating new analytics models essentially delivers major benefits. Actionable insights are generated by models like the Components Analysis and Recommendation Model, which promote productivity, cost savings, and sales growth. Utilizing the Components Analysis methodology, businesses can bargain for advantageous supplier discounts based on commonly used ingredients, cutting costs and expediting the procurement process. Similar to how effective tactics used by industry leaders like Pizza Hut, the Recommendation Model increases consumer engagement and boosts revenue by providing personalized pizza selections catered to individual tastes.

Moreover, the examination of well-known pizzas, such as the Classic Deluxe Pizza, highlights the need of menu flexibility in accommodating changing customer tastes. The report emphasizes that businesses may make dynamic adjustments to their offers through data-driven decision-making, which guarantees their competitiveness and relevance in a market that is changing quickly. Additionally, by offering insights into ingredient utilization and supplier negotiations, the Ingredients Analysis model helps to optimize operations, which eventually leads to higher profitability. To put it simply, food giants can remain ahead of the curve, boost sales, and improve operational efficiency by implementing advanced analytics models. Companies may efficiently navigate industry obstacles, meet customer needs, and sustain a competitive edge in the dynamic food market landscape by leveraging the power of data-driven insights.

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